
EECS 16A Designing Information Devices and Systems I

Homework 7

Submission Format

Your homework submission should consist of **one** file.

- `hw7.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned).

Submit the file to the appropriate assignment on Gradescope.

1. Reading Assignment

For this homework, please review Notes 12 through 14. Note 12 will provide an overview of voltage dividers and resistors. Note 13 will refresh you on how simple 1-D resistive touchscreens work, as well as the notion of power in electric circuits. Note 14 will cover a slightly more complicated 2-D resistive touchscreens and how to analyze them from a circuits perspective.

- Describe the significance of positive and negative power. How does this relate to passive sign convention?
- For a 2D resistive touchscreen, how would you measure horizontal position measurements and vertical position measurements?

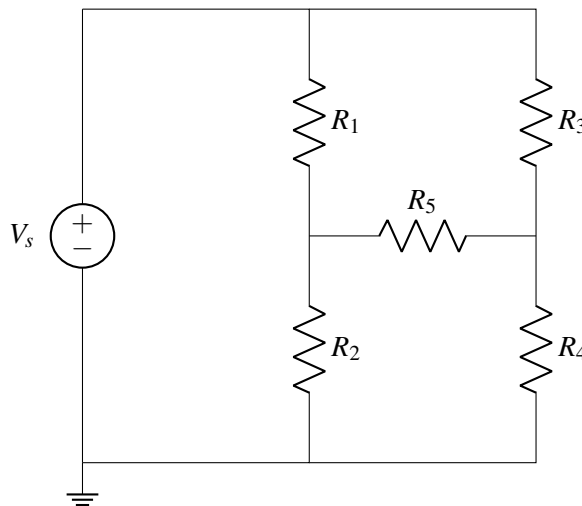
2. Circuit Analysis

(Contributors: Ava Tan, Aviral Pandey, Christos Adamopoulos, Matthew McPhail, Moses Won, Panos Zarkos, Raghav Anand, Titan Yuan, Urmita Sikder)

Learning Goal: This problem will help you practice circuit analysis using NVA method.

Using the steps outlined in lecture or in Note 11B, analyze the following circuit to calculate the currents through each element and the voltages at each node. Use the ground node labelled for you. You may use a numerical tool such as IPython to solve the final system of linear equations.

- $V_s = 5\text{ V}$, $R_1 = 1\ \Omega$, $R_2 = 2\ \Omega$, $R_3 = 3\ \Omega$, $R_4 = 4\ \Omega$, $R_5 = 5\ \Omega$



3. Resistive Touchscreen

(Contributors: Ava Tan, Ben Osoba, Christos Adamopoulos, Matthew McPhail, Moses Won, Panos Zarkos, Urmita Sikder, Wahid Rahman)

Learning Goal: The objective of this problem is to provide insight into modeling of resistive elements. This will also help to apply the concepts from 1D resistive touchscreen.

In this problem, we will investigate how a 1D resistive touchscreen with a defined thickness, width, and length can actually be modeled as a series combination of resistors. As we know the value of a resistor depends on its length.

Figure 1 shows the top view of a resistive touchscreen consisting of a conductive layer with resistivity ρ_1 , thickness t , width W , and length L . At the top and bottom it is connected through good conductors ($\rho = 0$) to the rest of the circuit. The touchscreen is wired to voltage source V_s .

Use the following numerical values in your calculations: $W = 50$ mm, $L = 80$ mm, $t = 1$ mm, $\rho_1 = 0.5 \Omega \text{m}$, $V_s = 5\text{V}$, $x_1 = 20$ mm, $x_2 = 45$ mm, $y_1 = 30$ mm, $y_2 = 60$ mm.

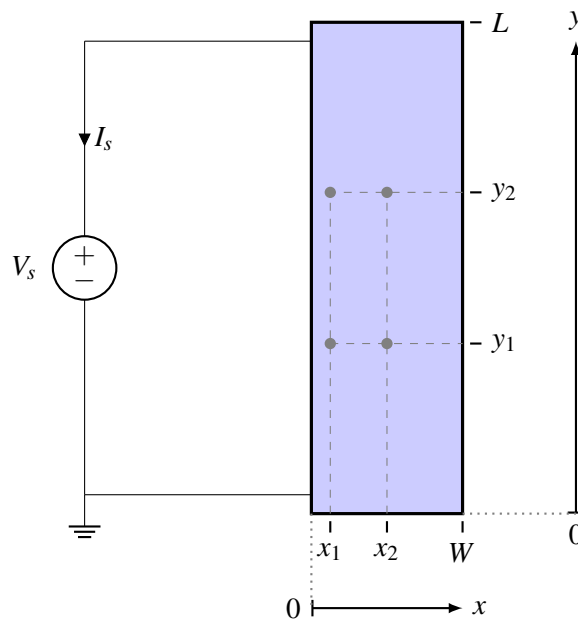
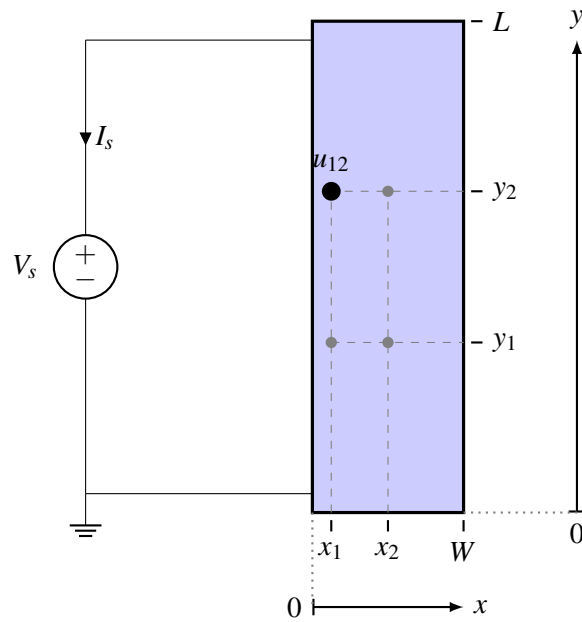


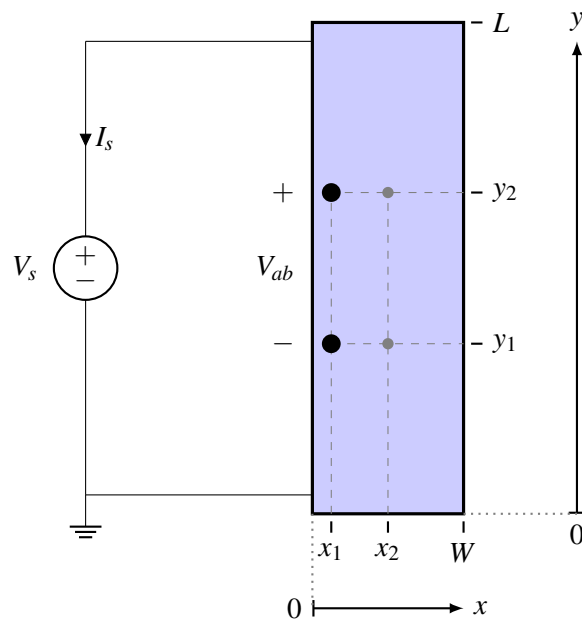
Figure 1: Top view of resistive touchscreen (not to scale). z axis i.e. the thickness not shown (into the page).

- Draw a circuit diagram representing Figure 1, where the touchscreen is represented as a resistor. **Note that no touch is occurring in this scenario.** Remember that circuit diagrams in general consist of only circuit elements (resistors, sources, etc) represented by symbols, connecting wires, and the reference/ground symbol. Calculate the value of current I_s based on the circuit diagram you drew. Do not forget to specify the correct unit as always.
- Let us assume u_{12} is the node voltage at the node represented by coordinates (x_1, y_2) of the touchscreen, as shown in Figure 2. What is the value of u_{12} ? You should first draw a circuit diagram representing Figure 2, which includes node u_{12} . Specify all resistance values in the diagram. Does the value of u_{12} change based on the value of the x -coordinate x_1 ?

Hint: You will need more than one resistor to represent this scenario.

Figure 2: Top view of resistive touchscreen showing node u_{12} .

- (c) Assume V_{ab} is the voltage measured between the nodes represented by touchscreen coordinates (x_1, y_1) and coordinates (x_1, y_2) , as shown in Figure 3. Calculate the absolute value of V_{ab} . As with the previous part, you should first draw the circuit diagram representing Figure 3, which includes V_{ab} . Calculate all resistor values in the circuit. *Hint: Try representing the segment of the touchscreen between these two coordinates as a separate resistor itself.*

Figure 3: Top view of resistive touchscreen showing voltage V_{ab} .

- (d) Calculate (the absolute value of) the voltage between the nodes represented by touchscreen coordinates (x_1, y_1) and coordinates (x_2, y_1) .
- (e) Calculate (the absolute value of) the voltage between the nodes represented by touchscreen coordinates (x_1, y_1) and coordinates (x_2, y_2) .

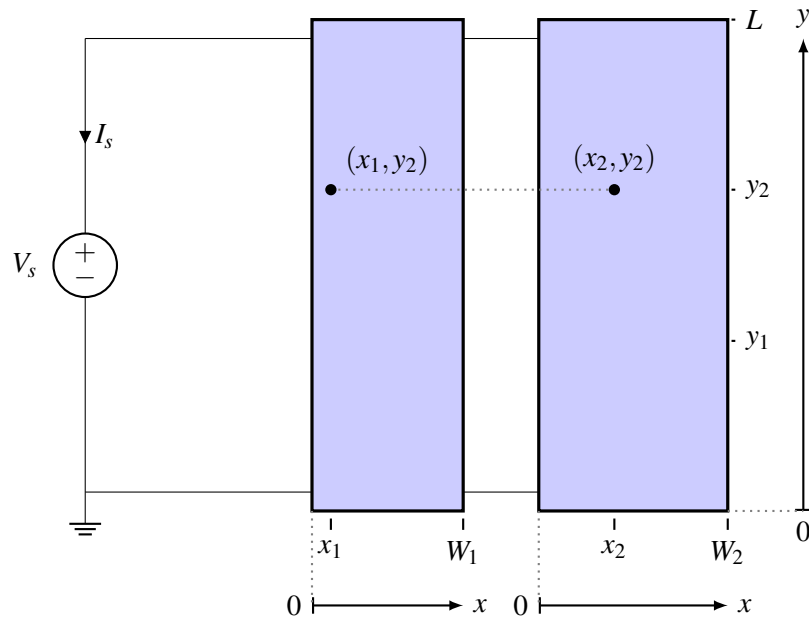


Figure 4: Top view of two touchscreens wired in parallel (not to scale). z axis not shown (into the page).

- (f) Figure 4 shows a new arrangement with two touchscreens. The second touchscreen (the one on the right) is identical to the one shown in Figure 1, except for different width, W_2 , and resistivity, ρ_2 .

Use the following numerical values in your calculations: $W_1 = 50$ mm, $L = 80$ mm, $t = 1$ mm, $\rho_1 = 0.5 \Omega\text{m}$, $V_s = 5\text{V}$, $x_1 = 20$ mm, $x_2 = 45$ mm, $y_1 = 30$ mm, $y_2 = 60$ mm, which are the same values as before. The new touchscreen has the following numerical values which are different: $W_2 = 85$ mm, $\rho_2 = 0.6 \Omega\text{m}$.

Draw a circuit diagram representing Figure 4, where the two touchscreens are represented as *two separate resistors*. **Note that no touch is occurring in this scenario.**

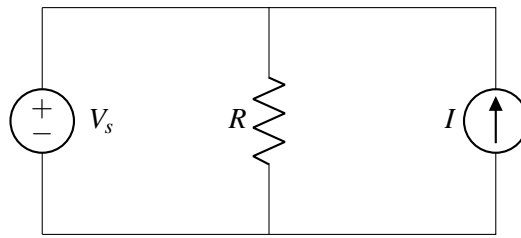
- (g) Calculate the value of current I_s for the two touchscreen arrangement based on the circuit diagram you drew in the last part.
- (h) Consider the two points: (x_1, y_2) in the touchscreen on the left, and (x_2, y_2) in the touchscreen on the right in Figure 4. Show that the node voltage at (x_1, y_2) is the same that at (x_2, y_2) , i.e. the potential difference between the two points is 0. You can show this without explicitly calculating the node voltages at the two points.

If you were to connect a wire between the two coordinates (x_1, y_2) in the touchscreen on the left, and (x_2, y_2) in the touchscreen on the right, would any current flow through this wire?

4. Power Analysis

(Contributors: Ava Tan, Craig Schindler, Moses Won, Raghav Anand, Urmita Sikder, Wahid Rahman)

Learning Goal: This problem aims to help you practice calculating power dissipation in different circuit elements. It will also give you insights into how power is conserved in a circuit.



- Find the expressions of power dissipated by each element in the circuit above. Remember to label voltage-current pairs using passive sign convention.
- Use $R = 5\text{k}\Omega$, $V_s = 5\text{V}$, and $I = 5\text{mA}$. Calculate the power dissipated by the voltage source (P_{V_s}), the current source (P_I), and the resistor (P_R).
- Once again, let $R = 5\text{k}\Omega$, $V_s = 5\text{V}$. What does the value I of the current source have to be such that the current source **dissipates** 40mW ? Note that it is possible for a current source to dissipate power, i.e. under passive sign convention, $P_I = 40\text{mW}$. For this value of I , compute I , P_{V_s} , P_I , and P_R as well.
As an aside: If the current source were delivering power it would have been $P_I = -40\text{mW}$, under passive sign convention, but this is NOT what the question is asking about.

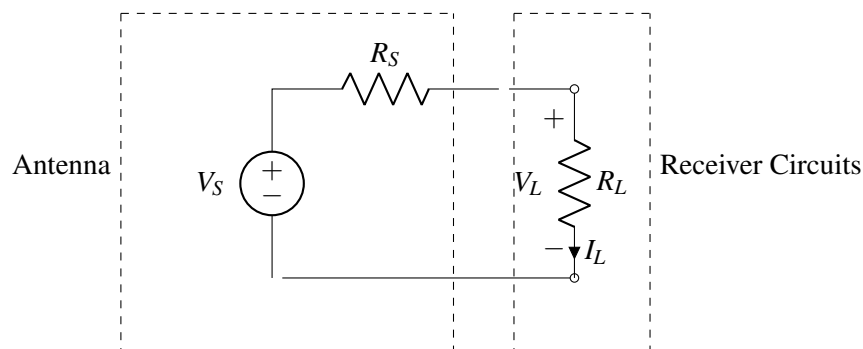
5. Maximum Power Transfer

(Contributors: Craig Schindler, Gireeja Ranade, Panos Zarkos, Urmita Sikder, Sashank Krishnamurthy)

Learning Goal: This problem shows how the power dissipated in the load depends on the value of the load resistance. It also helps to understand the condition required for maximum power transfer.

Smartphones use "bars" to indicate strength of the cellular signal. Fewer "bars" translate to slow or no connectivity. But what do these "bars" actually stand for? Voltage, current? Well, not quite. Good radio (a cellular modem is a particular type of radio) reception depends on the **power received at the receiver**. Communication theory tells us that higher received signal power enables higher data rates. To that end, we design a receiver that maximizes the power received, and hence connection speed.

A typical receiver consists of an antenna and receiver circuits. The antenna receives the radio waves propagating in space, and converts it into electrical voltages and currents. A very good abstraction used by circuit designers is to **model the antenna as a voltage source V_s , with a series resistance R_s** . The typical values of V_s in a real cellular receiver are in the range of micro- or milli-volts (10^{-6} and 10^{-3} , respectively) and the typical values of resistance R_s are usually 50Ω or 75Ω , depending on how the antenna is designed. The receiver circuits are quite complex and will be covered in detail in EE142 "Integrated Circuits for Communications". However, a standard abstraction is to **model these receiver circuits as a load resistance R_L** to the antenna, as shown in the figure below.



Models are very important in engineering design for their ability to abstract away details when they are not needed and are the key to successful design of complex systems. We will discuss the use and properties of electronic circuit models further in class.

Use the following component values for your calculations: $V_S = 100\mu V$, and $R_S = 50\Omega$.

Note: This homework problem and subsequent class content may require knowledge of calculus. For this problem, please review your understanding of how to compute derivatives.

- (a) Consider any value of R_L within the range: $0 \leq R_L \leq \infty$. Find the value of R_L that maximizes the **voltage** V_L across resistor R_L . Calculate the values of V_L , I_L , and the power P_L dissipated by resistor R_L for the value you found.
(Hint: The antenna voltage V_S and the resistance R_S are fixed. Intuitively argue for a particular value for R_L . How does the voltage change as the resistor value changes? Note: To rigorously prove your choice of R_L , you should express the voltage as a function of R_L , and differentiate it with respect to R_L .)
- (b) Consider any value of R_L within the range: $0 \leq R_L \leq \infty$. Find the value of R_L that maximizes the **current** I_L through resistor R_L . Calculate the values of V_L , I_L , and the power P_L dissipated by resistor R_L for the value you found.
(Hint: The antenna voltage V_S and the resistance R_S are fixed. Intuitively argue for a particular value of R_L . How does the current change as the resistor value changes?)
- (c) Find the value of R_L that maximizes the **power** P_L delivered to resistor R_L . Calculate the values of V_L , I_L , and the power P_L delivered to resistor R_L . **It is important to note that this value of R_L which maximizes the power delivered to R_L also optimizes cellular connectivity.** (Hint: The power optimization is best performed algebraically by setting the derivative of P_L with respect to R_L to 0. Alternatively you can do the optimization graphically. Refer to the code notebook provided and complete the code to plot P_L versus R_L and find the maximum.)

6. Hot Dogs - Fall 2018 Midterm 2

(Contributors: Alice Ye, Michael Kellman, Urmita Sikder)

Learning Goal: The objective of this problem is to explore how power is dissipated in different resistors in series/parallel combination.

Michael and Alice are going on a picnic with Daisy the robot dog! But when they arrive at the park, a thunderstorm starts with lots of lightning... However, Daisy still wants to play in the park!

- (a) We want to understand what will happen if Daisy is struck by lightning. A lightning bolt is represented with a voltage source of 1000 Volts (1 kV) with reference to the Earth. The lightning travels through the ionized air ($R_{Air} = 4k\Omega$), and through Daisy ($R_{Daisy} = 4k\Omega$) before reaching the ground. Daisy can withstand up to 10 Watts (W) of power before breakdown, (i.e. Daisy will breakdown if she dissipates >10 Watts).
Using the circuit diagram in Figure 5, find if it is safe for Daisy to be struck by a lightning bolt. Justify your answer by showing your work!

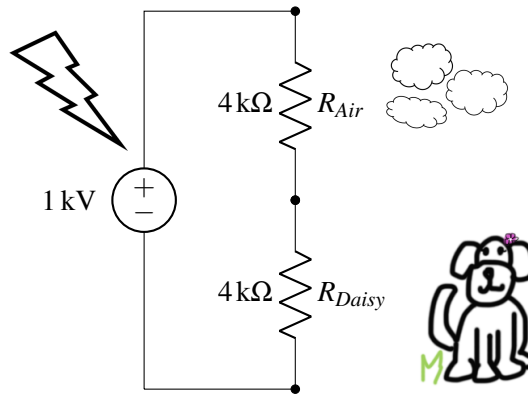


Figure 5: Circuit for Part (a)

- (b) Laura also brings along her dog, Elton, for the picnic. Elton has a resistance of $R_{Elton} = 2\text{ k}\Omega$ and can withstand up to 25 W of power. If Daisy and Elton hide together next to a tree of resistance $R_{Tree} = 4\text{ k}\Omega$ as shown in the Figure 6, would either of the dogs withstand being struck by lightning? Show your work.

Note: Resistors that have terminals connected to the same two nodes are said to be in parallel. In this case, R_{Tree} , R_{Daisy} , and R_{Elton} are parallel resistors. To find the equivalent resistance (R_{eq}) for n resistors in parallel, where $R_{eq} = R_1 || R_2 || R_3 || \dots || R_n$, compute the following: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$

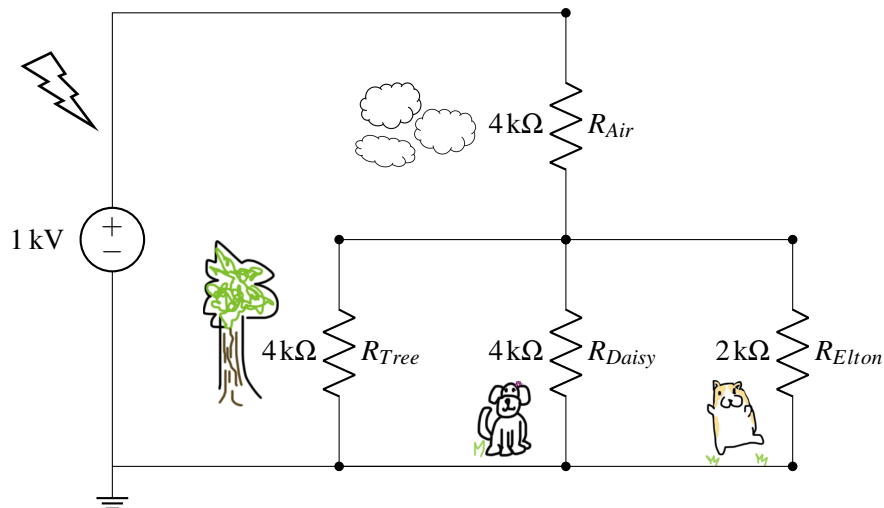


Figure 6: Circuit for Part (b)

7. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.